LAB 3

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Project: Inter-City Freight Load Matching Marketplace

Architecture Selection

“We chose Layered Architecture for the Inter-City Freight Load Matching Marketplace System.”

Architectural Choice:
We selected Layered Architecture, consisting of a Presentation Layer, Application/Business Logic Layer, and Data Layer.

Reason 1 (Separation of responsibilities):
The system performs several distinct operations. For example, when a shipper posts a shipment, the Presentation Layer collects details such as source, destination, cargo type, quantity, special handling requirements, and budget. The Application Layer then validates and processes these details for shipment matching, while the Data Layer stores the shipment information. Separating these responsibilities makes the system easier to develop and maintain.

Reason 2 (Supports different users and operations):
The marketplace has two major users: shippers and carriers, who perform different operations. For example, a shipper can post a shipment, view carrier bids, and select a carrier, while a carrier can view offered jobs and accept or reject them. A common Application Layer can handle these different operations while the Presentation Layer provides the appropriate interface to each user.

Security Advantage:
The layered structure allows authentication and verification to be handled before users access restricted marketplace functions. For example, an unverified carrier should not be allowed to submit a bid, and sensitive payment and proof-of-delivery data can be transmitted using encrypted protocols.

Performance Benefit:

The separation of layers allows the Application Layer to process operations such as shipment matching and bidding without the user interface directly accessing the database. This reduces unnecessary database interaction and allows the system to respond more efficiently to user requests.

UML:

